

Signal — Definitive Product Requirement Document (PRD) v2.0

Comprehensive Refinement: Security, Testing, Resilience, Flow & UI/UX Architecture

Project Code Name: Signal
Phase: MVP v1.0 → v1.5
Last Updated: May 2026
Status: Technical Specification Ready for Engineering Review

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1. Security Architecture & Threat Model

1.1 Threat Model (STRIDE Framework)

Threat Category	Asset	Attack Vector	Mitigation	Priority
Spoofting	User identity	OAuth token theft, session hijacking	PKCE + state param for OAuth; short-lived JWTs (15min); refresh token rotation; device fingerprinting	P0
Tampering	Workflow JSON, user data	MITM, supply chain	TLS 1.3 everywhere; signed webhooks (HMAC-SHA256); dependency pinning + SCA scanning (Snyk/Dependabot)	P0
Repudiation	Automated actions	User claims “I didn’t send that”	Immutable audit logs (WORM storage); digital signatures on all dispatched actions; non-repudiable consent records	P0

Table 1 – continued

Threat Category	Asset	Attack Vector	Mitigation	Priority
Information Disclosure	OAuth tokens, LLM prompts	DB breach, log leakage	AES-256-GCM encryption at rest (envelope encryption with KMS); token vault with zero-knowledge proxy; PII redaction in logs	P0
Denial of Service	LLM API, n8n engine	Token exhaustion, infinite loops	Rate limiting per user (token bucket); circuit breakers on LLM calls; execution timeouts (30s max); cost caps per tier	P1
Elevation of Privilege	YOLO Mode custom code	RCE via injected JS/Python	Sandboxed execution (gVisor/Firecracker); denylist for dangerous modules; AST analysis on injected code; no network egress from sandbox	P0

1.2 Authentication & Authorization Architecture

1.2.1 OAuth 2.0 + PKCE Implementation



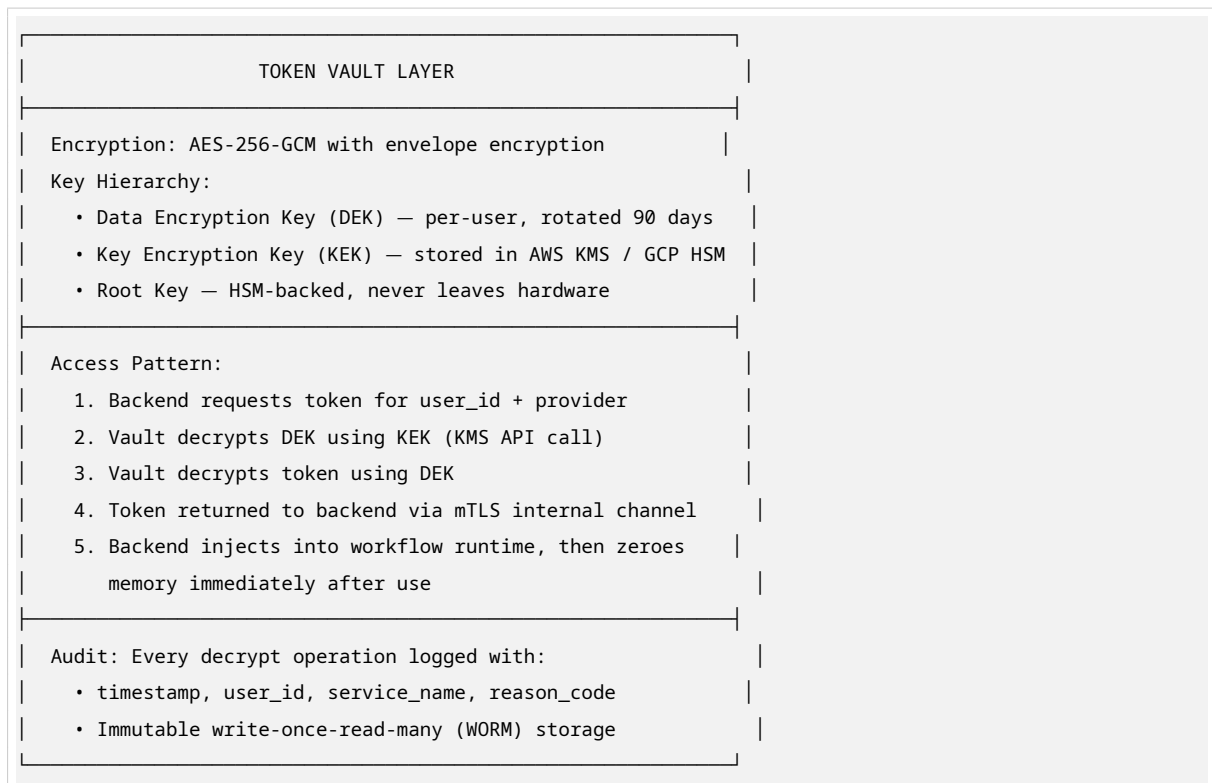
Specific Requirements:

- **PKCE (Proof Key for Code Exchange)** mandatory for all OAuth flows — prevents authoriza-

tion code interception attacks

- **State parameter** with CSRF nonce — 32-byte random, validated server-side, single-use, 10min TTL
- **Scope minimization** — request only scopes strictly needed. Gmail: <https://www.googleapis.com/auth/gmail.readonly> + <https://www.googleapis.com/auth/gmail.send> (send only for Smart Drafts). Never request <https://mail.google.com/> (full mailbox control).
- **Apple Sign-In** — enforce `response_mode=form_post` to prevent token leakage via URL fragments

1.2.2 Token Vault Architecture (Zero-Knowledge Proxy)



Critical Rules:

- OAuth tokens are **never** stored in application database plaintext
- Tokens are **never** logged (redacted via regex patterns in log pipeline)
- Tokens are **never** returned to frontend or exposed in API responses
- Token vault runs as **separate microservice** with its own IAM role and network isolation

1.2.3 Session Management

Property	Specification
Access Token	JWT, RS256 signed, 15-minute TTL
Refresh Token	Opaque string (UUIDv4), 7-day TTL, stored hashed (SHA-256) in DB
Rotation	Refresh token used once → new pair issued, old invalidated

Table 2 – continued

Property	Specification
Binding	Device fingerprint (hash of UA + IP subnet + screen res)
Revocation	Instant via Redis blocklist; webhook from OAuth provider triggers immediate invalidation
Idle Timeout	30 minutes (configurable per tier)
Absolute Timeout	7 days (forces re-auth)

1.2.4 Role-Based Access Control (RBAC)

Roles:

- user (default) — own data only
- admin (internal) — support view, read-only, with explicit user consent audit trail
- system — service-to-service, mTLS + SPIFFE identity

Permissions Matrix:

Resource	user	admin	system
Own workflows	CRUD	R	R
Own tokens	—	—	R*
Own digest content	R	R**	R
Other user data	—	—	—
System configs	—	—	CRUD

* system reads only from vault, never stores

** admin reads only with explicit user consent + ticket number

1.3 Data Protection & Privacy

1.3.1 Data Classification

Class	Examples	Storage	Retention	Encryption
Critical	OAuth tokens, API keys	Token Vault (HSM-backed)	Until revoked + 30 days	AES-256-GCM + envelope
Sensitive	Email content, chat logs, personal notes	Encrypted DB (per-user keys)	User-controlled + 90 days post-deletion	AES-256-GCM
Personal	Name, email, usage patterns	Standard DB	Account lifetime + 30 days	At-rest (AES-256)
Public	News articles, RSS content	Cached object storage	48 hours (deduplication cache)	TLS in transit only

1.3.2 LLM Data Handling

Zero-Retention Enforcement:

Provider	Zero-Retention Available	Implementation
Anthropic Claude	✅ Default (API data not used for training)	Use <code>anthropic-beta: messages-2023-12-15</code> API version
OpenAI GPT-4o	✅ Enterprise/Business tier only	Submit ZDR agreement; use <code>organization</code> header
Google Gemini	⚠️ Opt-out required	Set <code>x-goog-user-project</code> + disable data logging in console
Local Models (Llama 3)	✅ Fully private	Self-hosted on Signal infrastructure; data never leaves

Data Minimization for LLM Prompts:

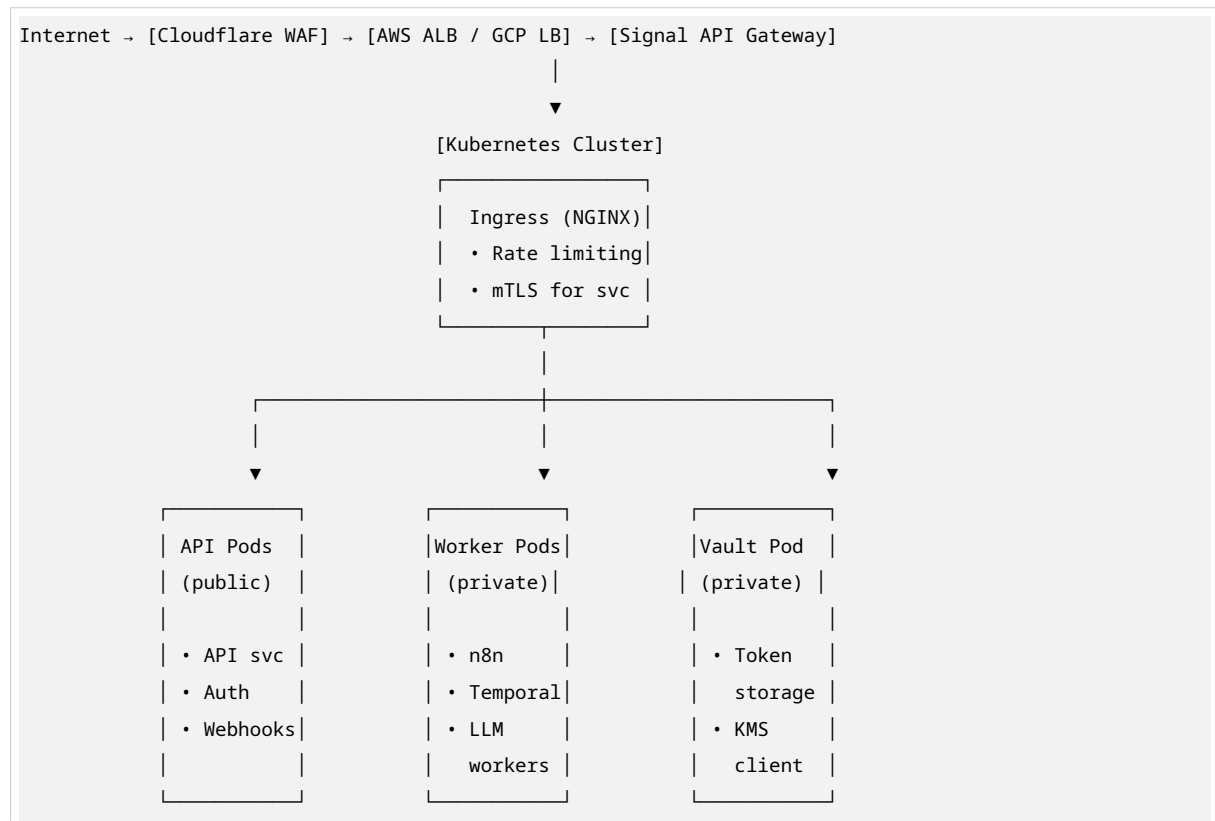
- **Pre-processing pipeline** strips PII before sending to LLM:
- Email addresses → [EMAIL_1], [EMAIL_2]
- Phone numbers → [PHONE_1]
- Names → [PERSON_1] (preserved for relationship context, but hashed)
- **Context window limits:** Never send >50 emails or >10,000 tokens per digest generation
- **No training data leakage:** System prompt explicitly instructs: “Do not memorize or reproduce user data. This is a stateless processing request.”

1.3.3 GDPR / CCPA Compliance

Requirement	Implementation
Right to Access	<code>/api/v1/user/data-export</code> —returns ZIP with all user data, all formats, within 30 seconds for Pro tier
Right to Deletion	<code>/api/v1/user/delete</code> —cascades through: token revocation → workflow deletion → vector DB purge → object storage purge → DB anonymization. 30-day grace period with recovery option.
Right to Portability	Export in JSON (machine-readable) + PDF (human-readable)
Consent Management	Granular consent per integration (Gmail, WhatsApp, etc.) with separate toggle + timestamp + version. Consent version bumps trigger re-prompt.
Data Processing Agreement	DPA signed with all subprocessors (OpenAI, Anthropic, cloud provider). Listed in <code>/privacy/subprocessors</code> .
Breach Notification	72-hour internal SLA; 72-hour regulatory notification; immediate user email if Critical/Sensitive class affected

1.4 Network Security

1.4.1 Infrastructure Hardening



Specific Controls:

- **WAF Rules:** OWASP Top 10 + custom rules for LLM prompt injection patterns (ignore previous instructions, DAN mode, jailbreak attempts)
- **Rate Limiting:** Token bucket per user_id + IP. Free: 10 req/min. Pro: 100 req/min. Business: 500 req/min. YOLO: 1000 req/min.
- **mTLS:** All internal service communication requires SPIFFE/SPIRE-issued certificates
- **Network Segmentation:** Worker pods (n8n, LLM) in private subnet with no public IP. NAT Gateway for outbound only.
- **Egress Filtering:** Worker pods can reach only: LLM APIs (allowlisted IPs), OAuth providers, email APIs. Default deny all other outbound.

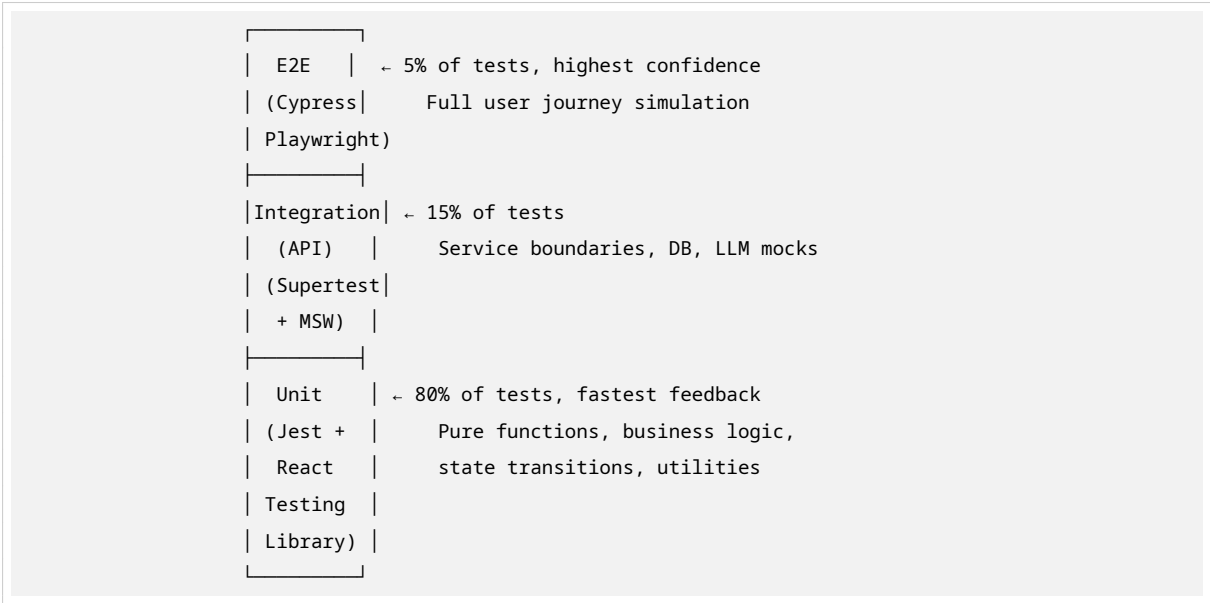
1.4.2 Webhook Security

All incoming webhooks (Gmail push notifications, WhatsApp Business API callbacks) must:

1. **Verify signature** using provider's public key (Google: X-Goog-Signature, Meta: SHA-256 signature)
2. **Idempotency check** — deduplicate via Message-Id or webhook payload hash (Redis 24h TTL)
3. **Replay protection** — reject timestamps >5 minutes old
4. **IP allowlisting** — only accept from known provider IP ranges

2. Testing Strategy & Quality Assurance

2.1 Testing Pyramid



Coverage Targets:

Layer	Target Coverage	Enforcement
Unit (business logic)	90%	CI gate –PR blocked if <90%
Unit (UI components)	70%	CI warning if <70%
Integration (API)	80%	CI gate –PR blocked if <80%
E2E (critical paths)	100% of P0 flows	CI gate –nightly + pre-release

2.2 Unit Testing Strategy

2.2.1 What to Test

Must Test (P0):

- Priority scoring algorithm (Critical Focus Block ranking)
- Deduplication engine (vector similarity + fuzzy matching)
- Token usage calculator (billing metering accuracy)
- PII redaction pipeline (regex + NER)
- Workflow schema validation (Signal-native JSON → backend translation)
- Consent state machine (grant → revoke → re-grant transitions)
- Rate limiter (token bucket algorithm edge cases)

Example: Priority Scoring Algorithm Test

```
// priority-engine.test.ts
describe('PriorityScoringEngine', () => {
  it('should rank unanswered client emails above internal newsletters', () => {
    const emails = [
      { sender: 'client@corp.com', daysUnanswered: 2, hasActionItem: true, domain: 'external' },
      { sender: 'newsletter@substack.com', daysUnanswered: 0, hasActionItem: false, domain: 'newsletter' }
    ];
    const ranked = scorePriority(emails);
    expect(ranked[0].sender).toBe('client@corp.com');
    expect(ranked[0].score).toBeGreaterThan(ranked[1].score);
  });

  it('should cap at 5 items regardless of input volume', () => {
    const emails = Array(50).fill({ sender: 'test@test.com', daysUnanswered: 1 });
    const ranked = scorePriority(emails);
    expect(ranked.length).toBe(5);
  });

  it('should handle empty input gracefully', () => {
    const ranked = scorePriority([]);
    expect(ranked).toEqual([]);
    expect(getEmptyStateMessage()).toBe('You are entirely caught up.');
```

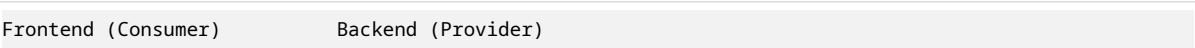
2.2.2 Mocking Strategy

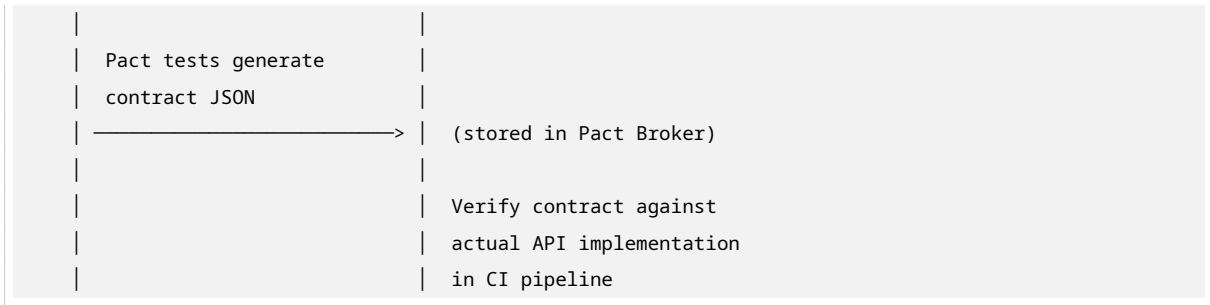
Dependency	Mock Tool	Strategy
LLM APIs (OpenAI/Anthropic)	msw (Mock Service Worker)	Return deterministic responses for test fixtures. Never hit real APIs in tests.
OAuth Providers	nock	Mock token exchange endpoints. Pre-canned token responses.
Database	testcontainers (PostgreSQL in Docker)	Spin up real DB for integration tests. Migrate schema. Seed with fixtures.
Vector DB	In-memory HNSW	Use lightweight in-memory vector index for deduplication tests.
Token Vault	jest.mock	Mock vault service with in-memory encrypted store.

2.3 Integration Testing Strategy

2.3.1 API Contract Testing

Use **Pact** (consumer-driven contract testing) to ensure frontend and backend compatibility:





Critical Contracts to Test:

1. GET /api/v1/digest — response shape, date formats, item limits
2. POST /api/v1/workflows — schema validation, error codes
3. POST /api/v1/drafts — smart draft generation, tone parameters
4. GET /api/v1/library/search — vector search results format

2.3.2 Workflow Integration Tests

Test complete data flows through the system:

```
Test: "Gmail → Deduplication → Summarization → Digest Delivery"

1. Inject test email fixture into mocked Gmail API
2. Trigger webhook (simulate Google push notification)
3. Verify: Email fetched → PII redacted → Deduplication checked →
   LLM mock called with correct prompt → Summary stored in DB →
   Digest API returns correct card format
4. Verify: Token vault decrypt called exactly once
5. Verify: Audit log entry created with correct action_type
```

2.3.3 LLM Integration Testing

Challenge: LLM outputs are non-deterministic.

Solution:

- Use **deterministic test mode** — mock LLM with structured JSON responses
- Separate **LLM prompt tests** (verify prompt construction) from **LLM output tests** (verify output parsing)
- Run **LLM evaluation suite** weekly with real APIs:
- 100 curated test inputs with “expected” output ranges
- Automated scoring via GPT-4-as-judge (consistency check)
- Regression detection — flag if output quality drops >10%

2.4 E2E Testing Strategy

2.4.1 Critical User Journeys (CUJs)

CUJ ID	Journey	Priority	Test Tool
CUJ-001	Sign up → Identity Box → Connect Gmail → View First Digest	P0	Playwright
CUJ-002	Receive email → Approve smart draft → Send → Verify sent	P0	Playwright + API verification
CUJ-003	Add RSS feed → Wait for digest → Verify deduplication	P1	Playwright
CUJ-004	Panic button (soft) → Verify data preserved → Reconnect	P1	Playwright
CUJ-005	Upgrade Free → Pro → Verify feature unlock + billing	P1	Playwright + Stripe test mode
CUJ-006	YOLO Mode: Inject custom LLM key → Run workflow → Verify cost attribution	P2	Playwright

2.4.2 E2E Test Environment

E2E Environment:

- Dedicated Kubernetes namespace: `signal-e2e`
- Real cloud services (test projects): Stripe test mode, OAuth test apps
- Database: Fresh PostgreSQL per test run (migrated + seeded)
- LLM: Mocked (deterministic) for speed; real API smoke test nightly
- Browser: Chromium + Firefox + WebKit (cross-browser matrix)
- Viewport: Desktop (1440x900), Tablet (768x1024), Mobile (375x812)

2.4.3 Visual Regression Testing

Use **Chromatic** or **Percy** for UI snapshot testing:

- Capture all core components in all states (empty, loading, error, populated)
- Capture full-page layouts at breakpoints (desktop, tablet, mobile)
- Baseline updates require design team approval
- Run on every PR that touches `/frontend/src/components`

2.5 Performance Testing

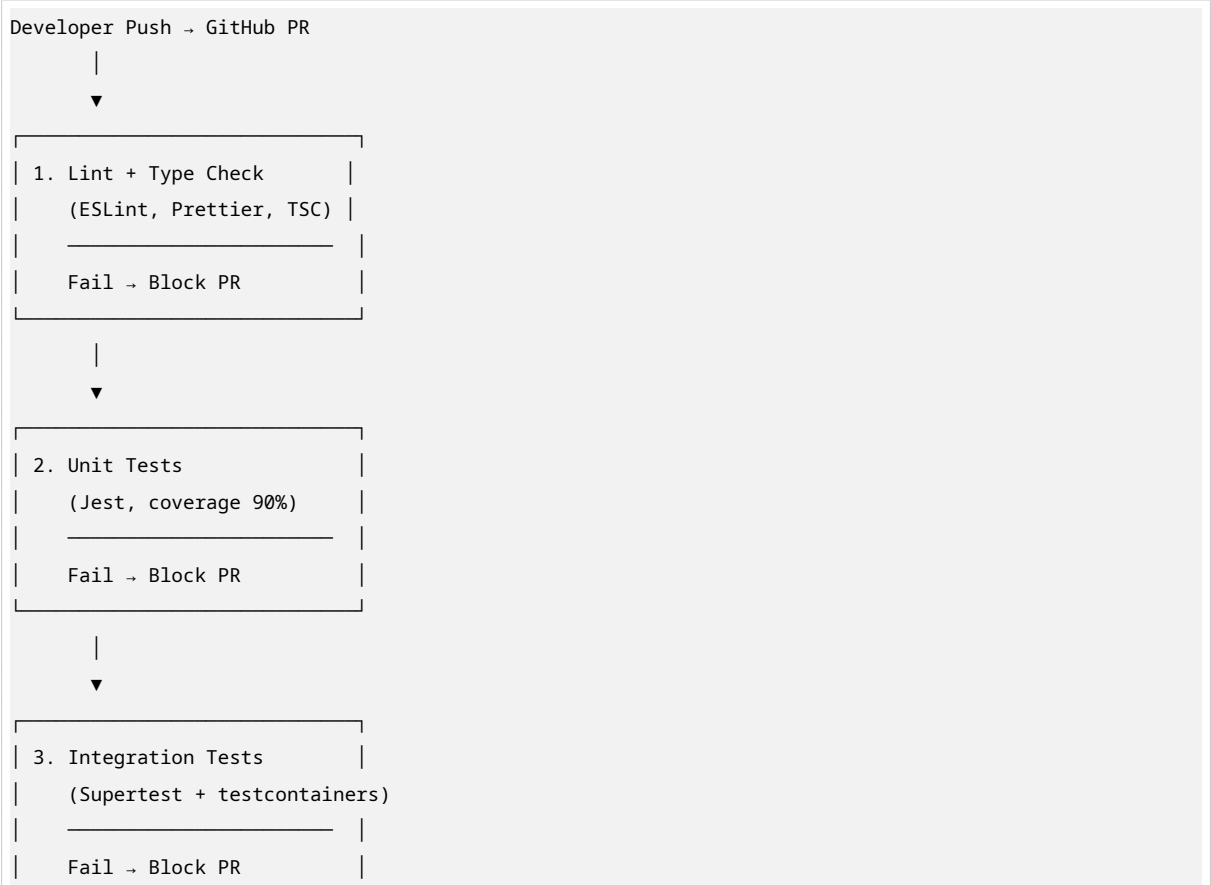
2.5.1 Load Testing Scenarios

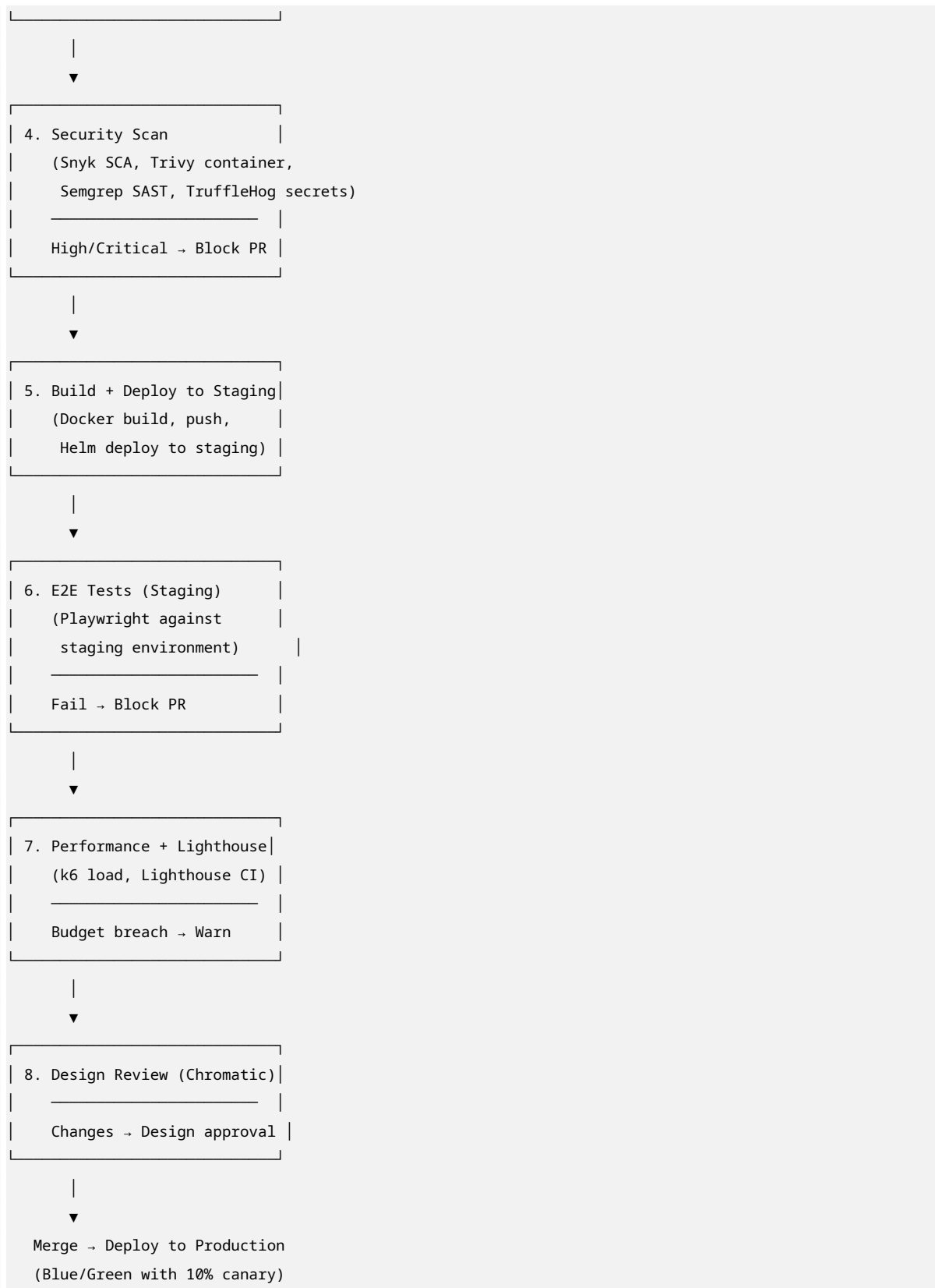
Scenario	Target	Tool	Frequency
Digest generation (1000 concurrent users)	p95 < 3s	k6	Weekly
API response time (auth, CRUD)	p95 < 200ms	k6	Weekly
Webhook ingestion burst (10k/min)	No dropped messages	k6 + custom	Monthly
LLM token throughput	1000 requests/min	Artillery	Monthly
Database query performance	Slow query log = 0	pgBadger	Daily (automated)

2.5.2 Performance Budgets

Metric	Budget	Enforcement
First Contentful Paint (FCP)	< 1.2s	Lighthouse CI gate
Largest Contentful Paint (LCP)	< 2.5s	Lighthouse CI gate
Time to Interactive (TTI)	< 3.5s	Lighthouse CI gate
Total Blocking Time (TBT)	< 200ms	Lighthouse CI gate
Cumulative Layout Shift (CLS)	< 0.1	Lighthouse CI gate
Bundle size (initial)	< 200KB gzipped	Bundle analyzer gate

2.6 CI/CD Pipeline





3. Resilience, Error Mitigation & Observability

3.1 Error Taxonomy & Handling Strategy

3.1.1 Error Classification

Code	Category	User Impact	System Action	AI Guide Message
E100	Transient (network blip, rate limit)	Brief delay	Retry 3× with exponential backoff (1s, 2s, 4s). Queue for later if persistent.	“I’ m catching up on your updates. One moment.”
E200	Integration (OAuth expired, API change)	Feature unavailable	Refresh OAuth token automatically. If fails, notify user to re-auth.	“Your Gmail connection needs a quick refresh. Tap here to reconnect — takes 5 seconds.”
E300	LLM Degradation (timeout, bad output)	Lower quality	Fallback to backup LLM (Claude → GPT-4o → local Llama 3). Flag for review.	“I summarized this slightly differently today. Let me know if it feels off.”
E400	User Input (invalid command, bad JSON)	Action blocked	Validate client-side + server-side. Return specific field errors.	“I didn’ t catch that. Try saying it like: ‘Draft a friendly reply to Joe.’”
E500	System (DB failure, crash)	Service degraded	Circuit breaker opens. Graceful degradation mode. PagerDuty alert.	“I’ m taking a quick breather. Your data is safe. I’ ll be back in a few minutes.”
E600	Security (auth failure, injection)	Session terminated	Immediate logout, revoke tokens, audit log, security team alert.	“I noticed something unusual. I’ ve signed you out to keep your account safe.”

3.1.2 Retry & Backoff Strategy

```
// Exponential backoff with jitter
async function resilientCall<T>({
  operation: () => Promise<T>,
  config: {
    maxRetries: number;
    baseDelayMs: number;
    maxDelayMs: number;
    retryableErrors: string[];
  }
}): Promise<T> {
  let lastError: Error;
```

```

for (let attempt = 0; attempt <= config.maxRetries; attempt++) {
  try {
    return await operation();
  } catch (error) {
    lastError = error;

    if (!config.retryableErrors.includes(error.code)) {
      throw error; // Non-retryable — fail fast
    }

    const delay = Math.min(
      config.baseDelayMs * Math.pow(2, attempt) + Math.random() * 1000,
      config.maxDelayMs
    );

    await sleep(delay);
  }
}

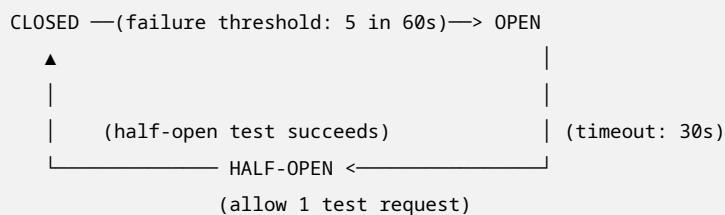
throw lastError; // All retries exhausted
}

// Usage for LLM calls
const summary = await resilientCall(
  () => llm.generateSummary(prompt),
  { maxRetries: 3, baseDelayMs: 1000, maxDelayMs: 10000, retryableErrors: ['E100', 'E300'] }
);

```

3.1.3 Circuit Breaker Pattern

State Machine:



Implementation for LLM API calls:

- **CLOSED:** Normal operation. Count failures per provider.
- **OPEN:** Stop sending requests to failing provider. Fallback to secondary provider immediately. Alert on-call.
- **HALF-OPEN:** After 30s, send 1 test request. If success → CLOSED. If failure → OPEN (extend timeout to 60s).

Implementation for n8n engine:

- If n8n instance crashes or returns 5xx >3 times in 5 minutes → circuit breaker OPEN

- Route new workflows to Temporal/custom micro-engine fallback
- Alert: “n8n shard [X] unhealthy — failover activated”

3.2 Graceful Degradation Matrix

Component	Full Function	Degraded Mode	Fallback Mode	User Notification
Smart Drafts	AI-generated, style-matched	Generic template draft	“Draft unavailable — compose manually”	Subtle chip: “Draft simplified”
Digest Generation	Full 24hr multi-source digest	Single-source digest (emails only)	“Digest delayed — check back in 1 hour”	Banner: “Running behind today”
Vector Search	Semantic + fuzzy matching	Keyword search only	“Search unavailable”	Chip: “Search mode: basic”
AI Guide Chat	Full conversational context	Last 5 messages context only	Static help links	Banner: “Chat history reset”
WhatsApp Integration	Bi-directional sync	Read-only sync	“WhatsApp disconnected”	Card: “Reconnect WhatsApp”
YOLO Mode	Custom JS/Python execution	Pre-built templates only	Disabled entirely	Modal: “Sandbox maintenance”

3.3 Observability Stack

3.3.1 Three Pillars

OBSERVABILITY STACK
METRICS (Prometheus + Grafana) • Business: signups, digests_generated, drafts_sent, mrr, churn_rate, nps_score • System: cpu/memory/disk per pod, db_connections, queue_depth, cache_hit_rate • LLM: tokens_per_request, latency_p50/p95/p99, cost_per_user, error_rate_by_provider • SLOs: digest_latency_p95 < 3s, api_availability > 99.9%
LOGS (Loki / Datadog) • Structured JSON logs with trace_id, span_id, user_id • Log levels: ERROR (alert), WARN (dashboard), INFO (analytics), DEBUG (on-demand sampling) • PII redaction pipeline: automatic before ingestion • Retention: 7 days hot, 30 days warm, 90 days cold
TRACES (Jaeger / Tempo) • OpenTelemetry instrumentation across all services • Trace every user request: frontend → API → worker → LLM → DB → response • Critical spans: digest_generation, draft_creation, oauth_refresh, vector_search

• Sampling: 100% for errors, 10% for success (configurable)|

3.3.2 Alerting Rules (PagerDuty)

Alert	Condition	Severity	Response
API Down	Error rate > 1% for 2 minutes	P1	Auto-page on-call engineer
LLM Latency Spike	p95 > 10s for 5 minutes	P2	Slack alert + auto-fallback to backup provider
OAuth Token Refresh Fail > 10 users	P2	Slack alert + user notification batch	
DB Connection Pool Exhausted	> 90% for 3 minutes	P1	Auto-scale DB connection pool + page DBA
n8n RCE Detected	Security alert triggered	P0	Immediate incident commander + auto-isolate pod + legal notification
Cost Anomaly	Daily LLM spend > 150% of 7-day avg	P2	Slack alert + billing freeze for flagged accounts
Churn Spike	Daily churn > 2x 30-day average	P3	Email to product team + auto-survey to churned users

3.3.3 User-Facing Health Dashboard

A public status page (status.signal.app) showing:

- Real-time system health (API, Digest Generation, LLM Providers, Integrations)
- Incident history with post-mortem links
- Planned maintenance windows
- Subscribe to SMS/email alerts for outages

3.4 Incident Response Playbook

3.4.1 Severity Definitions

Severity	Definition	Examples	Response Time
SEV-0	Complete outage, data breach, security incident	All users can't log in; OAuth tokens leaked; RCE exploited	5 minutes
SEV-1	Major feature degraded, revenue impact	Digest generation down for >50% users; payment processing failing	15 minutes

Table 14 – continued

Severity	Definition	Examples	Response Time
SEV-2	Minor feature degraded, workaround exists	Smart drafts failing for 1 provider; search slow	1 hour
SEV-3	Cosmetic, no user impact	Log noise; metric gaps; non-critical warnings	4 hours

3.4.2 Incident Response Runbook (SEV-0)

```
T+0min: Automated PagerDuty alert fires
T+2min: On-call engineer acknowledges, joins war room (Slack/Zoom)
T+5min: Incident Commander assigned (senior engineer or CTO)
T+5min: Auto-mitigation checks:
    • Circuit breakers opened?
    • Failover to backup region activated?
    • Affected users identified (auto-generated list)
T+10min: Public status page updated ("Investigating issue with X")
T+15min: Customer Success notified for proactive outreach to Business tier
T+30min: If unresolved, auto-escalate to secondary on-call + leadership
T+60min: Post-incident review scheduled (within 24 hours for SEV-0)
T+24h: Post-mortem published (internally) + status page update
T+7d: Remediation tickets created and prioritized
```

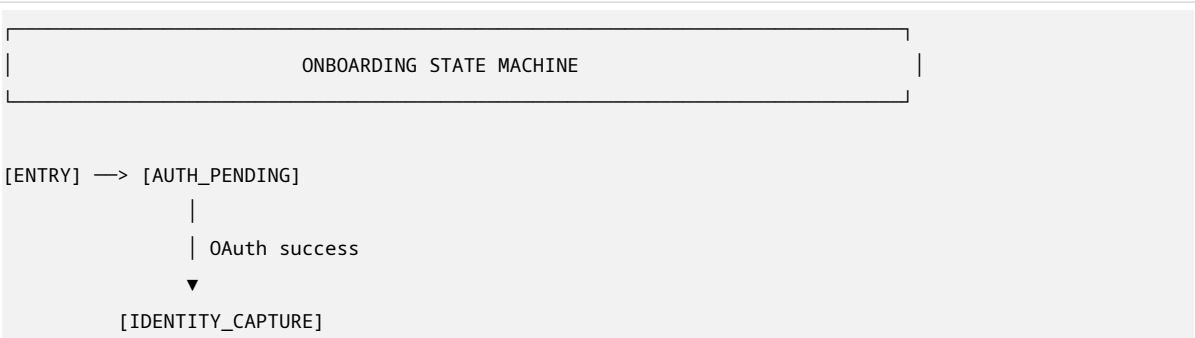
3.4.3 Chaos Engineering

Monthly **Game Days** to test resilience:

- 1. **Kill n8n pod** — verify failover to Temporal/custom engine
- 2. **Throttle LLM API** — verify circuit breaker + fallback provider
- 3. **Corrupt 10% of DB connections** — verify connection pool recovery
- 4. **Expire OAuth tokens for 50 test users** — verify auto-refresh + graceful degradation
- 5. **Simulate DDoS** — verify WAF + rate limiting + origin protection

4. User Flow Architecture & State Machines

4.1 Complete Onboarding Flow (State Machine)



```

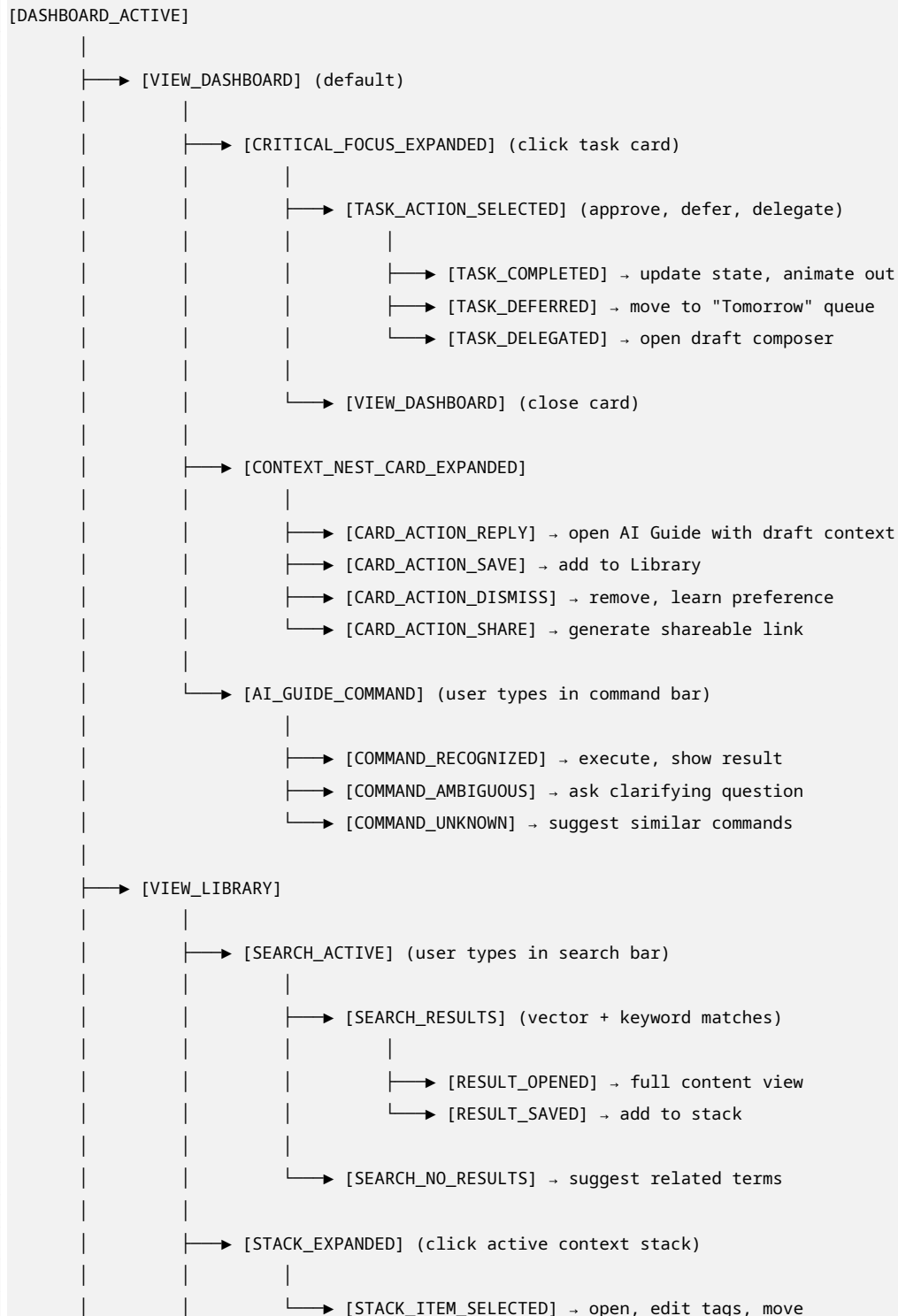
|
| User submits Identity Box
| System: NLP parse → persona tag → template selection
▼
[PERSONA_ASSIGNED]
|
| System: Suggest integrations (progressive auth)
▼
[INTEGRATION_AUTH_1] (e.g., Gmail)
|
| Auth success OR skip
| If success: Test connection (ping API)
▼
[INTEGRATION_TEST_1]
|
| Connection valid?
|   YES → [INTEGRATION_AUTH_2] (optional, e.g., WhatsApp)
|   NO  → [INTEGRATION_RETRY_1] (max 3 attempts)
▼
[INTEGRATION_AUTH_2] —(same pattern)—> [INTEGRATION_TEST_2]
|
| All integrations complete OR user skips
▼
[DATA_FETCH]
|
| System: Fetch last 24h data from connected sources
|         (async, with progress bar: "Reading your emails...")
|         Timeout: 45 seconds max
▼
[DIGEST_GENERATION]
|
| LLM processing + deduplication
|         Fallback: Show raw data if LLM fails
▼
[FIRST_DIGEST_READY]
|
| Display digest inline (the "Instant Win")
|         If empty: "You're entirely caught up! Here's what I'll watch for."
▼
[TOMORROW_PREVIEW]
|
| System: "Tomorrow at 6 AM, I'll deliver your Morning Brief
|         with [X] emails summarized and [Y] action items."
▼
[ONBOARDING_COMPLETE]
|
| Redirect to Dashboard
|         Trigger: Welcome email + in-app tutorial tooltip sequence
▼
[DASHBOARD_ACTIVE]

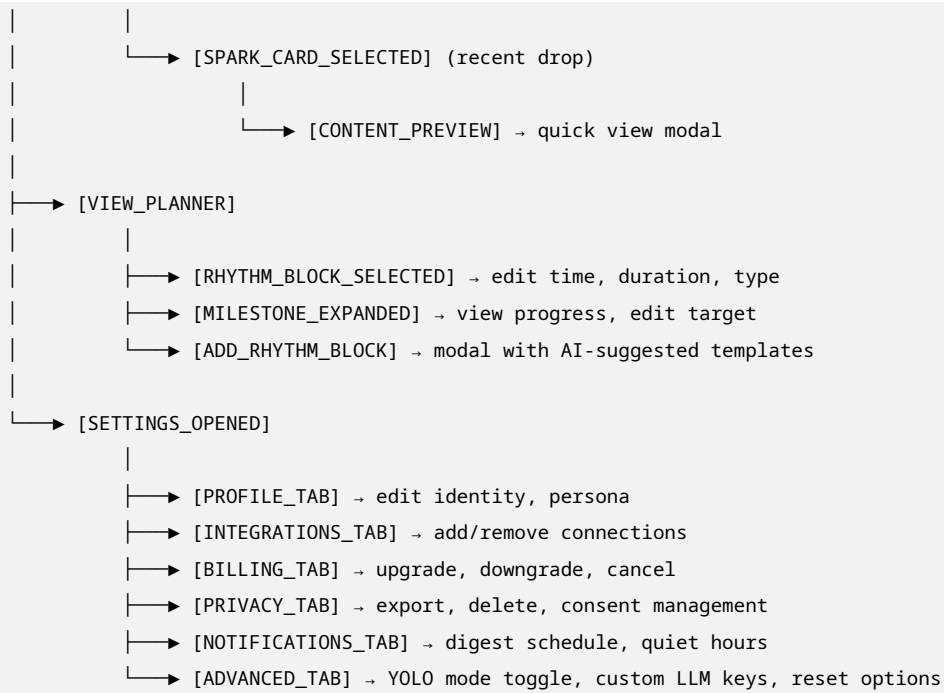
```

EDGE CASES:

- [AUTH_PENDING] → timeout 10min → show "Having trouble? Use email magic link instead"
- [IDENTITY_CAPTURE] → empty input → allow skip, assign "General Professional" persona
- [INTEGRATION_AUTH_1] → user declines all → enter "Manual Mode" (no integrations, library-only)
- [DATA_FETCH] → timeout 45s → show partial data + "Still catching up on the rest..."
- [DIGEST_GENERATION] → LLM failure → show raw email subjects + "I need a moment to summarize these"
- [FIRST_DIGEST_READY] → user dismisses without interacting → trigger email with digest 1 hour later

4.2 Core Dashboard State Machine

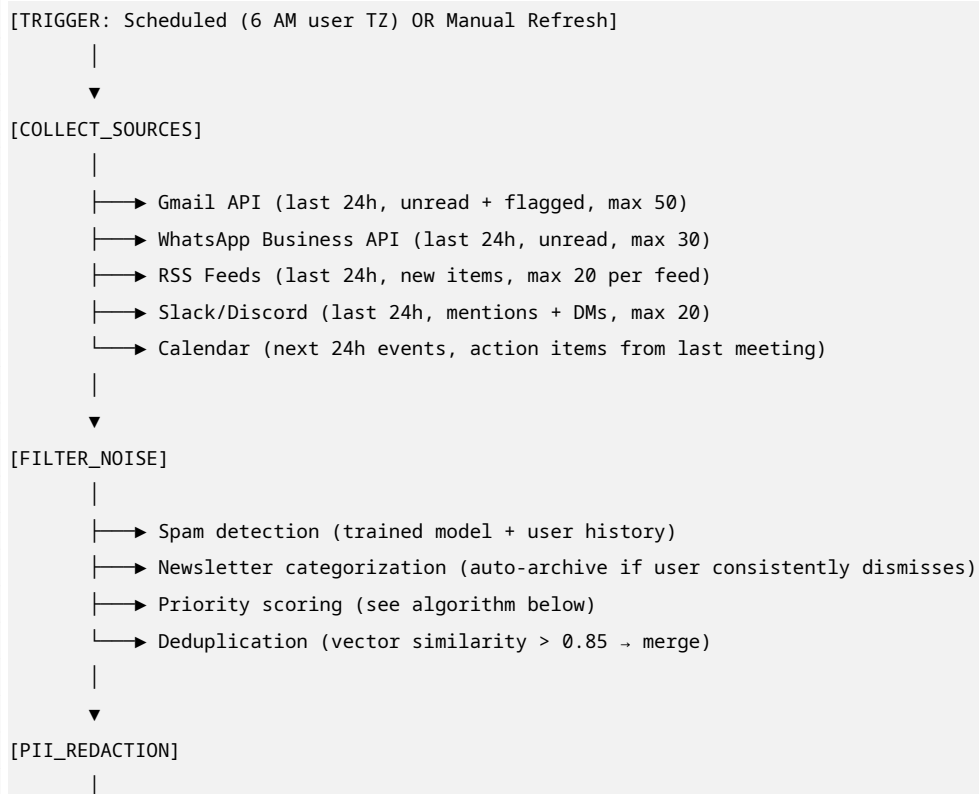


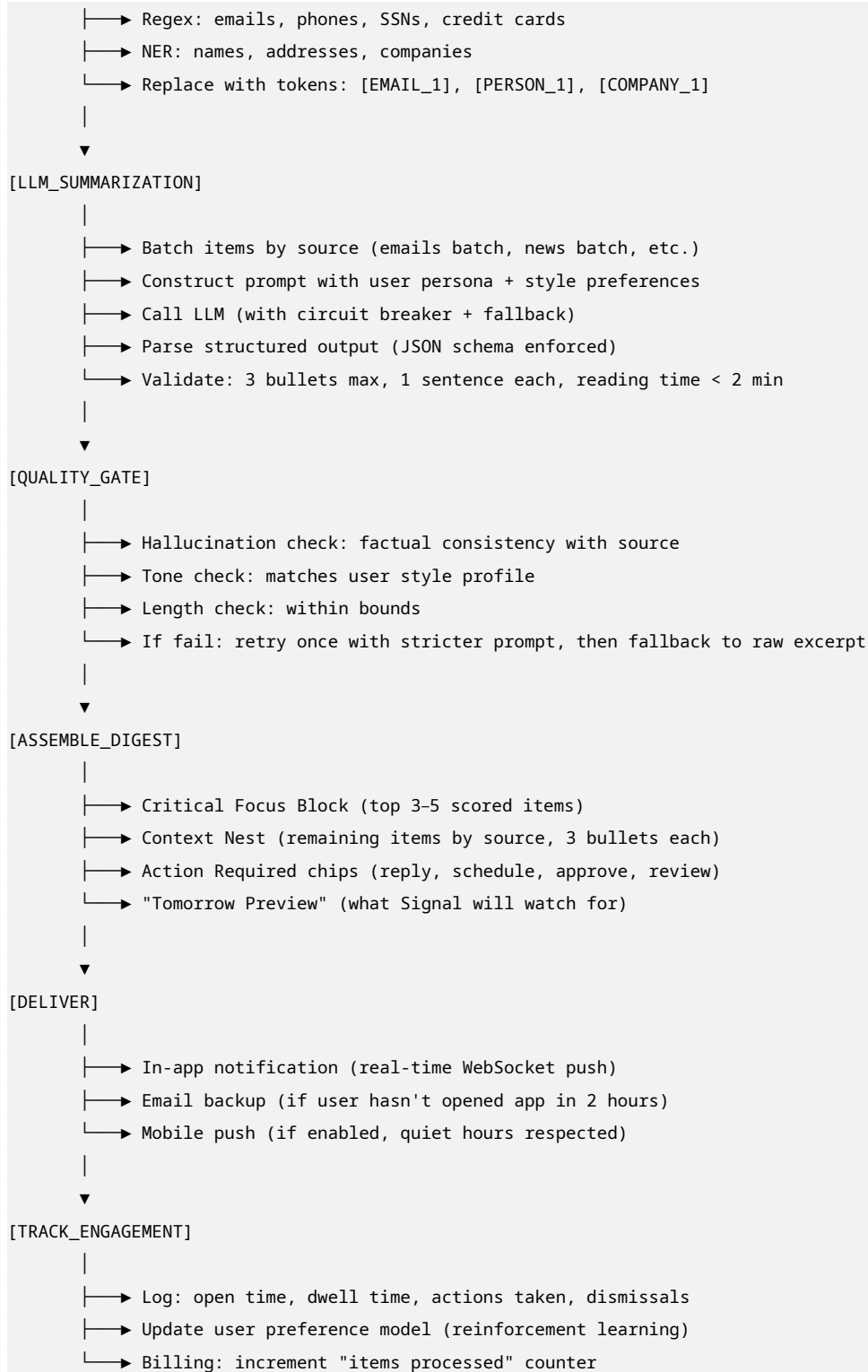


GLOBAL STATES (available from any view):

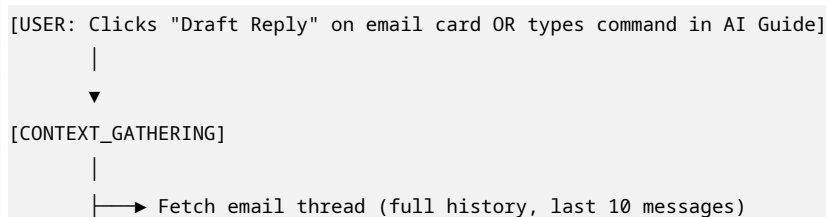
- [PARK_DISTRACTION] → quick capture modal → dumps to inbox → AI Guide acknowledges
- [PANIC_BUTTON_SOFT] → confirmation modal → flush cache, reset AI context
- [PANIC_BUTTON_HARD] → multi-step confirmation + export offer → full wipe
- [HELP_OVERLAY] → contextual tips based on current view

4.3 Digest Generation Flow (Backend)





4.4 Smart Draft Flow





4.5 Edge Case Catalog

Scenario	System Behavior	User Experience
User has zero emails in 24h	Skip email section, show “No new emails” with celebration animation	“You’ re entirely caught up.”
LLM returns malformed JSON	Retry once with stricter schema prompt. If still bad, fallback to raw excerpt + manual note.	Card shows raw subject + “I couldn’ t summarize this one neatly.”
OAuth token revoked by user externally	Detect on next API call (401). Auto-trigger re-auth flow.	Banner: “Gmail needs reconnecting. Tap here.”
User hits usage cap mid-month	Queue new items but don’ t process. Show upgrade prompt.	“You’ ve reached your calm limit. Upgrade for unlimited peace of mind.”

Table 15 – continued

Scenario	System Behavior	User Experience
Two emails are 90% similar	Deduplicate into single card with “+1 similar” badge.	Single card: “2 emails about project deadline”
User types gibberish in command bar	Intent classification fails. Suggest 3 most likely commands.	“Did you mean: ‘Draft reply to Joe’ or ‘Summarize last meeting’?”
n8n workflow enters infinite loop	Execution timeout (30s). Circuit breaker opens. Kill container.	“This automation took too long. I’ve paused it for your safety.”
User deletes account during active digest generation	Cancel all jobs for user_id. Queue cleanup tasks. Show “Deletion in progress…”	“We’re securely wiping your data. This takes up to 30 seconds.”
Mobile device rotated during onboarding	Persist state. Resume at current step. No data loss.	Seamless continuation.
User has 200+ unread emails	Process top 50 by priority. Queue rest for next digest.	“I prioritized the 50 most important emails. The rest are in your queue.”

5. UI/UX Design System & Interaction Patterns

5.1 Design Philosophy: “Digital Minimalism”

Core Principles:

- 1. **Cognitive Load Budget:** Every screen element must earn its place. If it doesn’t reduce decisions or provide clarity, remove it.
- 2. **Progressive Disclosure:** Show the minimum viable information. Reveal depth only on intent (hover, click, expand).
- 3. **Calm Technology:** No blinking, no bouncing, no red alerts. Status changes are subtle (color shifts, gentle fade-ins).
- 4. **Friction by Design:** Destructive actions (send, delete, reset) require intentional confirmation. Helpful actions (save, draft) are one-tap.

5.2 Design Tokens

5.2.1 Color Palette

```
:root {
  /* Base */
  --color-bg-primary: #FAF8F8;      /* Warm off-white — paper-like calm */
  --color-bg-secondary: #F5F5F0;    /* Slightly darker for cards/panels */
  --color-bg-tertiary: #EBE5E5;     /* Borders, dividers */
  --color-bg-inverse: #1A1A1A;      /* Dark mode primary */

  /* Text */
```

```

--color-text-primary: #1A1A1A;    /* High contrast for reading */
--color-text-secondary: #6B6B6B;  /* Metadata, timestamps */
--color-text-tertiary: #9E9E9E;   /* Disabled, hints */
--color-text-inverse: #FAFAF8;    /* On dark backgrounds */

/* Accent — Single accent color system for consistency */
--color-accent: #2D6A4F;          /* Deep forest green — growth, calm, action */
--color-accent-light: #40916C;    /* Hover states */
--color-accent-dark: #1B4332;     /* Active states */
--color-accent-muted: #D8F3DC;    /* Backgrounds for accent-related items */

/* Semantic */
--color-success: #2D6A4F;         /* Same as accent — positive is the default */
--color-warning: #D4A373;         /* Warm amber — gentle attention */
--color-error: #C75146;           /* Muted red — serious but not alarming */
--color-info: #74C69D;           /* Light green — neutral information */

/* Dark Mode Overrides */
--dark-bg-primary: #121212;
--dark-bg-secondary: #1E1E1E;
--dark-text-primary: #E8E8E8;
--dark-accent: #52B788;
}

```

Color Usage Rules:

- **Never use pure black (#000000) or pure white (#FFFFFF).** Always use the warm off-whites and soft blacks above.
- **Accent color is used sparingly:** Primary buttons, active states, progress indicators, and the “Approve” action.
- **Error red is never used for non-critical states.** A failed draft uses the warning amber, not red. Red is reserved for account security and deletion.

5.2.2 Typography

```

:root {
  /* Font Stack */
  --font-sans: 'Inter', -apple-system, BlinkMacSystemFont, 'Segoe UI', sans-serif;
  --font-mono: 'JetBrains Mono', 'SF Mono', monospace;

  /* Scale — Major Third (1.25) ratio, base 16px */
  --text-xs: 0.75rem;    /* 12px — captions, badges */
  --text-sm: 0.875rem;   /* 14px — secondary text, metadata */
  --text-base: 1rem;     /* 16px — body, cards */
  --text-lg: 1.25rem;    /* 20px — subheadings, command bar */
  --text-xl: 1.5rem;     /* 24px — section headers */
  --text-2xl: 1.875rem;  /* 30px — page titles */
  --text-3xl: 2.25rem;   /* 36px — hero, onboarding headlines */

  /* Line Height */

```

```

--leading-tight: 1.25; /* Headings */
--leading-normal: 1.5; /* Body text */
--leading-relaxed: 1.75; /* Long-form reading (Library) */

/* Weight */
--weight-normal: 400;
--weight-medium: 500; /* Labels, buttons */
--weight-semibold: 600; /* Headings, emphasis */
}

```

Typography Rules:

- **Maximum 2 font weights per screen.** Usually 400 (body) + 600 (headings).
- **Card titles:** 16px semibold, 1.25 line height, max 2 lines (truncate with ellipsis).
- **Bullet points:** 14px normal, 1.5 line height, max 1 line per bullet (strictly enforced).
- **Command bar input:** 16px medium, placeholder in tertiary color.

5.2.3 Spacing & Layout

```

:root {
  /* 4px base grid */
  --space-1: 0.25rem; /* 4px — icon padding */
  --space-2: 0.5rem; /* 8px — tight gaps */
  --space-3: 0.75rem; /* 12px — card internal padding */
  --space-4: 1rem; /* 16px — standard gap */
  --space-5: 1.25rem; /* 20px — section separation */
  --space-6: 1.5rem; /* 24px — card external margin */
  --space-8: 2rem; /* 32px — panel padding */
  --space-10: 2.5rem; /* 40px — major section breaks */
  --space-12: 3rem; /* 48px — page margins */

  /* Border Radius */
  --radius-sm: 4px; /* Buttons, badges */
  --radius-md: 8px; /* Cards, inputs */
  --radius-lg: 12px; /* Modals, panels */
  --radius-xl: 16px; /* Onboarding containers */
  --radius-full: 9999px; /* Pills, avatars */

  /* Shadows — Subtle, never harsh */
  --shadow-sm: 0 1px 2px rgba(26, 26, 26, 0.04);
  --shadow-md: 0 4px 12px rgba(26, 26, 26, 0.06);
  --shadow-lg: 0 8px 24px rgba(26, 26, 26, 0.08);
  --shadow-focus: 0 0 0 3px rgba(45, 106, 79, 0.15); /* Accent ring */
}

```

5.2.4 Motion & Animation

```

:root {
  /* Durations */
  --duration-instant: 100ms; /* Hover states, color changes */
}

```

```
--duration-fast: 200ms;      /* Button presses, toggles */
--duration-normal: 300ms;    /* Card transitions, panel slides */
--duration-slow: 500ms;      /* Page transitions, modal entrances */

/* Easing */
--ease-default: cubic-bezier(0.4, 0, 0.2, 1);    /* Standard — smooth */
--ease-decelerate: cubic-bezier(0, 0, 0.2, 1);    /* Entering — soft landing */
--ease-accelerate: cubic-bezier(0.4, 0, 1, 1);    /* Exiting — quick departure */
--ease-bounce: cubic-bezier(0.34, 1.56, 0.64, 1); /* Playful — limited use */
}
```

Animation Rules:

- **No animation > 500ms.** User attention is precious.
- **Respect prefers-reduced-motion:** All animations degrade to instant transitions.
- **Purposeful motion only:** Elements move to show relationship (hierarchy, causality), never for decoration.
- **Loading states:** Never use spinners for > 500ms. Use skeleton screens or progressive text reveal (“Reading emails...” → “Finding priorities...” → “Almost there...”).

5.3 Component Library

5.3.1 Core Components

Card (Context Nest)

[icon] Source Name [time] 2h

- First bullet point here, one line max
- Second bullet point, also one line
- Third bullet point, final summary

[Reply] [Save] [Dismiss]

← 12px secondary, top row

← 14px normal, bullet list

← Action chips, 12px medium

Specs:

- Background: --color-bg-secondary
- Border: 1px solid --color-bg-tertiary
- Border-radius: --radius-md (8px)
- Padding: --space-3 (12px)
- Margin-bottom: --space-4 (16px)
- Hover: shadow-md transition, 200ms
- Max-height: 140px (enforces 3-bullet constraint)
- Focus ring: shadow-focus (for keyboard navigation)

Critical Focus Block

⚡ Critical Focus [3 of 5]

← Accent color top border (2px)

[1] Reply to Client about contract	← Numbered, 16px semibold
Due: Today • From: Gmail	← 12px secondary metadata
[Draft Reply] [Defer to Tomorrow]	← Primary + secondary buttons
[2] Review Q3 budget proposal	
Due: Tomorrow • From: Slack	
[Open] [Remind me at 3pm]	
[3] Call with investor rescheduled	
Due: Today • From: Calendar	
[View Calendar] [Propose New Time]	

Specs:

- Top border: 2px solid --color-accent
- Background: --color-bg-primary (slightly different from nest cards)
- Numbering: 20px semibold, accent color, circular background
- Empty state: Centered illustration + "You are entirely caught up." + subtle confetti animation

AI Guide Panel (Right 30%)

 Signal Guide	← Sticky header, 16px semibold
Status: "I processed 142 items of noise for you today. Relax."	← 14px secondary, updates every 30 minutes or on major event
Contextual Actions: [Draft Reply] [Schedule] [Save]	← Dynamic button row Changes based on active card
— Chat History —	
Guide: "I noticed 3 emails from clients waiting. Want me to draft responses?"	← 14px normal, max 50 messages retained (scrollable)
User: "Draft friendly reply to Joe saying Thursday works"	
Guide: "Done. Here's the draft: [preview card] [Approve] [Edit]"	
[Park a Distraction]	← Quick action, always visible
Type a command...	← Input field, 16px medium
  	← Voice, attach, quick actions

- Width: 30% desktop, min 320px, max 400px
- Background: --color-bg-secondary
- Border-left: 1px solid --color-bg-tertiary
- Height: 100vh, sticky position
- Chat bubbles: User right-aligned (accent-muted bg), Guide left-aligned (bg-primary)
- Input: Fixed bottom, auto-resize textarea (max 5 lines)
- "Park a Distraction": Floating button above input, accent color, pulse animation on new thought

Input States:

- Empty: Placeholder "Type a command or ask me anything..."
- Typing: Real-time intent suggestion dropdown (max 5 suggestions)
- Recognized: Green left border + "I'll draft a reply to Joe"
- Ambiguous: Yellow left border + "Did you mean reply to Joe or schedule with Joe?"
- Unknown: Red left border + "I'm not sure. Try: 'Draft reply to [name]'"

- Trigger: "/" prefix for slash commands (e.g., /draft, /schedule, /search)
- Trigger: "@" prefix for mentioning contacts (pulls from relationship graph)
- Trigger: "#" prefix for tags (pulls from Library tags)

“Park a Distraction” Modal

Trigger: Click floating button OR keyboard shortcut (Ctrl/Cmd + Shift + D)

The diagram illustrates a distraction management system. It features a central box with a dashed border. At the top left of this box is a red brain icon followed by the text "Park a Distraction". Below this is a horizontal line. In the center of the box is the text "Dump it here. I'll hold it so you can focus." Below this text is another horizontal line. Underneath the line is a smaller rectangular box containing the text "I need to remember to call the plumber about the office leak...". To the right of this smaller box, outside the main dashed box, is the text "Auto-resize textarea" and "No formatting, plain text". At the bottom of the main dashed box are two buttons: "[Save & Return to Focus]" and "[Cancel]". Below the main dashed box is the text "Tip: I'll remind you about this during your next Rest block."

- Escape key → close without saving

- Enter key → save (no multi-line needed for quick thoughts)
- Shift+Enter → new line (for longer thoughts)
- Saved items appear in "Distraction Inbox" in Library
- Auto-tagged: "parked-[timestamp]"

Panic Button Flow (Soft Reset)

Step 1: User clicks Settings → Advanced → "Reset Workspace"

Step 2: Confirmation Modal:

 Reset Workspace?

This will:
Clear your current dashboard view
Reset AI Guide conversation history
Flush temporary cache
Keep all your connected apps
Keep all saved Library items
Keep your account and billing

[Export my data first]
[Reset Workspace — I understand]
[Cancel]

Step 3: Progress indicator (2 seconds):

"Clearing workspace... → Resetting AI context... → Done."

Step 4: Success state:

Dashboard reloads with fresh state. AI Guide says: "Fresh start. What should we focus on today?"

Panic Button Flow (Hard Reset)

Step 1: User clicks Settings → Advanced → "Delete Everything"

Step 2: Warning Modal (irreversible action):

 Delete All Data?

This is permanent and cannot be
undone. We will:
• Delete all workflows and automations
• Revoke all connected app permissions
• Remove all saved Library items
• Delete your account and billing info
• Cancel any active subscriptions

[Download my data (ZIP)]

Step 3: Type-to-Confirm (friction by design):

"Type 'DELETE MY ACCOUNT' to confirm: [_____]"

- Button disabled until exact match
- 3-second cooldown after typing match before button enables

Step 4: Final confirmation:

"Are you absolutely sure? [Yes, delete everything] [No, keep my account]"

Step 5: Execution (async, up to 30 seconds):

Progress: "Revoking OAuth tokens... → Deleting workflows... → Removing data... → Goodbye."

Step 6: Redirect to landing page with message: "Your account has been deleted. We'd love to have you
↩ back."

5.3.3 Empty States

Location	Empty State	Visual	Action
Critical Focus	“You are entirely caught up.”	Subtle animated checkmark + soft confetti	“Set a milestone for today?”
Context Nest	“No new updates from [Source].”	Source icon in muted gray	“Adjust source settings”
Library	“Your knowledge library is empty.”	Open book illustration	“Save your first article” (with demo)
Search Results	“No matches for ‘[query]’”	Magnifying glass + question mark	“Try: [suggested related term]”
Planner	“Today’ s rhythm is open.”	Sunrise illustration	“Add your first focus block”
AI Guide (first use)	“Hi, I’ m your Signal Guide.”	Friendly avatar wave animation	“What should I know about you?”

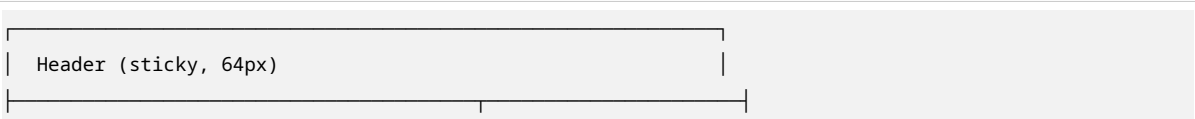
5.4 Responsive Behavior

5.4.1 Breakpoint System

```
/* Mobile First */
--breakpoint-sm: 640px; /* Large phones, small tablets */
--breakpoint-md: 768px; /* Tablets */
--breakpoint-lg: 1024px; /* Small laptops */
--breakpoint-xl: 1280px; /* Desktops */
--breakpoint-2xl: 1536px; /* Large monitors */
```

5.4.2 Layout Transformations

Desktop (1280px): 70/30 Split



Panel A: Clarity Board (70%)	Panel B: AI Guide (30%)
• Critical Focus	• Status Banner
• Context Nest	• Contextual Actions
• View switcher (Dashboard/Library/Planner)	• Chat History
	• Command Bar

Tablet (768–1279px): Collapsible Guide

Header + [Guide Toggle] (hamburger icon)
Panel A: Clarity Board (100%) • Critical Focus • Context Nest • View switcher [Swipe from right edge OR tap Guide Toggle] ▼ AI Guide (slide-over, 400px wide, overlay) • Same content as desktop

Mobile (<768px): Stacked with Bottom Sheet

Header (48px, compact)
Critical Focus (full width, max 3 items, swipeable cards)
Context Nest (vertical scroll, cards stack, full width)
[View Switcher: Dashboard Library Planner] (tab bar, bottom sticky)
[🧠 Guide] [💬 Chat] [⚡ Command] ← Bottom nav

Guide Interaction:

- Tap [🗨️ Guide] → Bottom sheet slides up (70% height)
- Tap [💬 Chat] → Full-screen chat view
- Tap [⚡ Command] → Inline command bar (sticky bottom, replaces nav)

5.4.3 Touch & Gesture Patterns

Gesture	Action	Feedback
Swipe right on card	Quick-save to Library	Card slides right, green checkmark appears, haptic feedback (light)
Swipe left on card	Dismiss / Mark read	Card slides left, red “X” appears, haptic feedback (medium)
Long press on card	Peek preview	Card elevates (shadow-lg), preview modal appears
Pull down on dashboard	Refresh digest	Loading indicator, fetch new data
Pinch on Library card	Zoom to full content	Smooth scale transition to detail view
Two-finger tap on AI Guide	“Park a Distraction” quick open	Modal appears instantly

5.5 Accessibility (a11y) Specification

5.5.1 WCAG 2.1 AA Compliance Targets

Criterion	Implementation	Verification
1.1.1 Non-text Content	All icons have <code>aria-label</code> or <code>aria-hidden</code> + adjacent text. Images have descriptive alt text.	axe-core scan
1.3.1 Info & Relationships	Semantic HTML: <code><header></code> , <code><main></code> , <code><nav></code> , <code><article></code> , <code><aside></code> . ARIA landmarks for panel regions.	Screen reader test (NVDA, VoiceOver)
1.4.3 Contrast	All text 4.5:1 ratio. Large text 3:1. Interactive elements 3:1.	Color contrast analyzer
1.4.10 Reflow	Layout works at 320px width without horizontal scroll.	Browser resize test
1.4.11 Non-text Contrast	UI components (cards, buttons, inputs) have 3:1 contrast against adjacent colors.	Visual inspection
2.1.1 Keyboard	All functionality available via keyboard. Tab order logical. No keyboard traps.	Manual keyboard navigation test
2.2.2 Pause/Stop/Hide	Auto-updating content (status banner) can be paused. No auto-playing audio/video.	axe-core scan
2.4.3 Focus Order	Focus moves predictably: top-to-bottom, left-to-right.	Tab-through test

Table 18 – continued

Criterion	Implementation	Verification
2.4.7 Focus Visible	All interactive elements have visible focus ring (<code>shadow-focus</code> token).	Visual inspection
4.1.2 Name/Role/Value	All custom components expose correct ARIA roles, states, and properties.	axe-core + screen reader

5.5.2 Screen Reader Experience

Dashboard Announcement (on load):

"Signal Dashboard. 3 critical items waiting.
12 new emails summarized. 2 news stories.
AI Guide panel available."

Card Announcement:

"Email from Client Corp. 3 bullet points.
First: Contract renewal due tomorrow.
Second: Budget approval pending.
Third: Meeting request for Thursday.
Actions available: Reply, Save, Dismiss."

AI Guide Response:

"Signal Guide says: I drafted a friendly reply to Joe.
Preview: 'Hi Joe, Thursday works perfectly for me...'
Actions: Approve and send, Make friendlier, Make firmer, Edit."

Empty State Announcement:

"You are entirely caught up. No critical items.
All data streams clear."

5.5.3 Keyboard Shortcuts

Shortcut	Action	Context
?	Open keyboard shortcut help	Global
j / k	Navigate next / previous card	Dashboard
Enter	Open selected card / Approve action	Dashboard / Guide
e	Quick-reply to selected email	Dashboard (email card focused)
s	Save to Library	Dashboard (any card focused)
d	Dismiss / Mark read	Dashboard (any card focused)
Ctrl/Cmd + Shift + D	Park a Distraction	Global
Ctrl/Cmd + K	Focus command bar	Global
Ctrl/Cmd + 1/2/3	Switch view: Dashboard / Library / Planner	Global

Table 19 – continued

Shortcut	Action	Context
Escape	Close modal / Cancel action / Blur input	Global
Shift + R	Refresh digest	Dashboard

5.5.4 Motion & Vestibular Disorders

```
@media (prefers-reduced-motion: reduce) {
  * {
    animation-duration: 0.01ms !important;
    animation-iteration-count: 1 !important;
    transition-duration: 0.01ms !important;
  }

  /* Replace animated loading with static text */
  .loading-skeleton::after {
    content: "Loading...";
    animation: none;
  }

  /* Disable parallax, zoom effects */
  .card-hover-zoom {
    transform: none;
  }
}
```

5.6 Micro-interactions & Feedback

5.6.1 Success States

Action	Visual Feedback	Audio/Haptic
Approve & Send draft	Card slides out (accelerate ease), green flash at send destination, count “Sent” increments	Haptic: Success (light)
Save to Library	Card compresses slightly, green checkmark morphs from save icon, “Saved” toast (2s)	Haptic: Light tap
Dismiss card	Card shrinks + fades (200ms), remaining cards slide up to fill space	Haptic: Medium tap
Complete all Critical Focus	Confetti particles (subtle, 20 max, 1s duration), “You are entirely caught up” fades in	None (calm celebration)
Digest ready	Gentle pulse on notification dot, AI Guide header glows briefly	None (no interruption)

5.6.2 Loading States

Duration	Pattern	Example
< 200ms	Instant —no loading state	Button press, toggle
200ms –500ms	Skeleton screen (shimmer)	Card content loading
500ms –2s	Progressive text reveal	“Reading emails…” → “Finding priorities…” → “Almost there…”
> 2s	Progress bar + cancel option	Large data import, initial onboarding
Indeterminate	Animated ellipsis + context	“Signal is thinking…”

5.6.3 Error States (User-Facing)

Error	UI Treatment	Recovery Action
Network offline	Banner: “You’ re offline. Signal will sync when you’ re back.” + grayed-out UI	Auto-retry when connection restored. Queue actions.
Draft generation failed	Card shows: “I couldn’ t draft this one. [Try again] [Compose manually]”	One-click retry with fallback provider.
OAuth disconnected	Card replaced with: “[Source] needs reconnecting. [Reconnect]”	Deep-link to OAuth re-auth flow.
Usage limit reached	Banner: “You’ ve reached your monthly limit. [Upgrade]”	Soft block —read-only mode. No data loss.
LLM hallucination detected	Card flagged: “I summarized this, but please double-check. [View original]”	Confidence score < 70% triggers this.

5.7 Dark Mode Specification

5.7.1 Color Inversion Rules

```
Light Mode → Dark Mode Mapping:
--color-bg-primary (#FAF8F8) → --dark-bg-primary (#121212)
--color-bg-secondary (#F5F5F0) → --dark-bg-secondary (#1E1E1E)
--color-text-primary (#1A1A1A) → --dark-text-primary (#E8E8E8)
--color-accent (#2D6A4F) → --dark-accent (#52B788) [lighter for contrast]
--color-error (#C75146) → #E07A5F [lighter, more visible]
--shadow-md → 0 4px 12px rgba(0, 0, 0, 0.3) [stronger shadows for depth]
```

5.7.2 Dark Mode Specifics

- **Images:** All user-uploaded images get `brightness(0.9)` filter. News article thumbnails unchanged.
- **Charts/Graphs:** Invert data colors for milestones. Use `--dark-accent` for positive trends.
- **Code blocks (YOLO Mode):** Use `atom-one-dark` or similar theme. Never pure white background.
- **Transitions:** Switching light/dark triggers 300ms color transition on all themed elements.

- **Default:** Respect `prefers-color-scheme`. Allow manual override in Settings. Persist choice in `localStorage`.

6. Appendix: Decision Log & Risk Matrix

6.1 Architecture Decision Log (ADL)

ID	Date	Decision	Context	Alternatives Considered	Status
ADL-001	2026-05	Use n8n as optional backend, not sole dependency	Licensing risk, multi-tenancy gaps	Temporal, Prefect, custom engine	Pending — requires n8n enterprise license negotiation
ADL-002	2026-05	Signal-native workflow schema	Decouple from n8n JSON, enable provider switching	Direct n8n JSON exposure	Approved
ADL-003	2026-05	Token Vault with zero-knowledge proxy	Security, compliance, user trust	HashiCorp Vault, AWS Secrets Manager	Approved — custom build for cost control
ADL-004	2026-05	Anthropic Claude as default LLM	Stronger privacy posture (no training by default), better long-context	OpenAI GPT-4o, Google Gemini, local Llama	Approved — with OpenAI fallback for vision tasks
ADL-005	2026-05	Freemium model (Free/Pro/Business/YOLO)	Market anchor at \$20, conversion optimization	7-day trial only, 2-tier structure	Approved — replaces original trial model
ADL-006	2026-05	React + TypeScript frontend	Team expertise, ecosystem, performance	Vue, Svelte, vanilla JS	Approved
ADL-007	2026-05	PostgreSQL + Redis + Vector DB (Pinecone)	Relational data, caching, semantic search	MongoDB, Supabase, Chroma	Approved
ADL-008	2026-05	Kubernetes + Helm deployment	Scalability, industry standard, cost	Serverless (AWS Lambda), PaaS (Heroku)	Approved — with serverless for sporadic workloads

6.2 Risk Matrix (Updated)

Risk	Likelihood (1-5)	Impact (1-5)	Risk Score	Mitigation Owner	Status
n8n license revocation	3	5	15	CTO / Legal	Active — enterprise contact initiated
n8n critical security vulnerability	4	5	20	Security Lead	Active —sandbox isolation, WAF rules
LLM API cost spike	3	4	12	Engineering Lead	Active —caching layer, cost caps
User data privacy breach	2	5	10	Security Lead	Active — encryption, ZDR, SOC 2 roadmap
Competitor launches “Zen” product	3	3	9	Product Lead	Monitoring — speed to market critical
Pricing resistance at \$29 Pro tier	3	3	9	Growth Lead	Testing —A/B test \$24 vs \$29 vs \$39
YOLO Mode RCE via user code	4	5	20	Security Lead	Active —gVisor sandbox, AST analysis
OAuth provider API changes	3	3	9	Engineering Lead	Monitoring — abstraction layer reduces impact
Team scaling bottleneck	3	4	12	CTO	Planning —hiring roadmap for Q3
LLM output quality regression	4	3	12	ML Lead	Active —weekly eval suite, prompt versioning

6.3 Glossary

Term	Definition
Critical Focus Block	Dashboard panel showing top 3–5 highest-priority actionable items
Context Nest	Grid of cards containing compressed data streams (emails, news, chats)
Identity Box	Onboarding input field for user to describe their work and overwhelm
Instant Win	First live digest generated during onboarding to prove immediate value
Park a Distraction	Quick-capture feature to offload thoughts without breaking focus

Table 25 – continued

Term	Definition
Signal-native schema	Internal JSON format for workflows, decoupled from backend engine specifics
Spark Card	Recently added item in Library (email forward, browser extension save, etc.)
YOLO Mode	Advanced tier allowing custom LLM keys, raw workflow JSON, and guardrail disabling
Zero-Knowledge Proxy	Token vault architecture where Signal backend cannot read plaintext API keys
Zero-Retention API	LLM provider agreement guaranteeing user data is not used for model training

Document Control

Version	Date	Author	Changes
1.0	2026-05	Product Team	Initial PRD
2.0	2026-05	Research + Engineering	Added Security, Testing, Resilience, Flow, UI/UX sections

Next Review Date: 2026-06-15
Distribution: Core team, investors, engineering leads, security auditor